

Education

2017	Ph.D. Statistics	The University of British Columbia - Vancouver, BC
2012	M.Sc. Statistics	Brock University - St. Catharines, ON
2011	B.Sc. Mathematics	Brock University - St. Catharines, ON
2011	B.Sc. Biology (3-year)	Brock University - St. Catharines, ON

Employment History

2025-2027	Research Fellow	Politecnico di Milano, Italy
	➤ Advances in the statistical modelling of risk due to compound events, with application in flooding under climate change.	
2022-2025	Senior Statistician	BGC Engineering Inc. - Canada
	➤ Statistical modelling of complex earth science applications like flooding and climate change.	
2020-2022	Assistant Professor of Teaching	The University of British Columbia, Vancouver
	Department of Statistics	
	➤ Taught and developed courses at all levels in statistics and data science.	
2019-2020	Lecturer	The University of British Columbia, Vancouver
	Master of Data Science Program and Dept. of Statistics	
	➤ Taught and developed courses in the Master of Data Science program.	
2017-2019	Postdoctoral Teaching and Learning Fellow	The University of British Columbia, Vancouver
	Master of Data Science Program and Dept. of Statistics	
	➤ Taught and developed courses in the Master of Data Science program.	

Publications

1. Abdelmoaty, H. M., Papalexiou, S. M., Mamalakis, A., Singh, S., Coia, V., Hairabedian, M., Szeftel, P., & Grover, P. (2025). Generative adversarial networks for downscaling hourly precipitation in the canadian prairies [Under minor revision]. *Journal of Geophysical Research: Machine Learning and Computation*.
2. LeSueur, P., Zubrycky, S., McDougall, S., Coia, V., Strouth, A., & Porter, M. (2025). Vulnerability of people in wood-frame buildings to rapid flow-type landslides [Conditionally accepted]. *Natural Hazards*.
3. Bale, S., Coia, V., Russell, B., & Clohan, D. (2024). *Probability distributions of tailings dam breach volumes by failure mode as part of a risk screening-level tool*.
4. Coia, V., Joe, H., & Nolde, N. (2024). Copula-based conditional tail indices. *Journal of Multivariate Analysis*, 201, 105268.
5. Vidal-Llana, X., Salort Sánchez, C., Coia, V., & Guillen, M. (2023). Non-crossing dual neural network: Joint value at risk and conditional tail expectation regression with non-crossing conditions. *Available at SSRN 4351877*.
6. Coia, V., Joe, H., & Nolde, N. (2021). Tail behavior for bivariate distributions based on pareto mixtures. *Advances in Statistics-Theory and Applications: Honoring the Contributions of Barry C. Arnold in Statistical Science*, 207–228.
7. Coia, V. (2017). *Forecasting of nonlinear extreme quantiles using copula models* [PhD thesis]. University of British Columbia.
8. Coia, V., & Huang, M. L. (2014). A sieve model for extreme values. *Journal of Statistical Computation and Simulation*, 84(8), 1692–1710.
9. Ayad, M., Coia, V., & Kihel, O. (2014). The number of relatively prime subsets of a finite union of sets of consecutive integers. *J. Integer Seq.*, 17(3), 14–13.
10. Huang, M. L., Coia, V., & Brill, P. (2013). A cluster truncated pareto distribution and its applications. *International Scholarly Research Notices*, 2013.
11. Coia, V. (2012). On estimation of extreme value distributions. *Brock Report in Mathematics and Statistics*, 120809-01.

Software (R Packages)

1. Coia, V., Joshi, A., Tan, S., & Zhu, Z. (2024). *Distionary: Create and evaluate univariate probability distributions*. <https://distionary.netlify.app/>
2. Coia, V., Joshi, A., Tan, S., & Zhu, Z. (2024). *Distplyr: Manipulation of univariate distributions*. <https://distplyr.netlify.app/>
3. Coia, V. (2024). *Famish: Refine a family of distributions to match observations*. <https://probaverse.github.io/famish/>
4. Coia, V., Wang, J., MacKenzie, L., Eaton, B., & Davidson, S. (2024). *Sxchan: Simple channel cross sections*. <https://stochagbem.github.io/sxchan/>
5. Eaton, B., Davidson, S., MacKenzie, L., & Coia, V. (2024). *Gbem: Gravel bed river bank erosion model*. <https://stochagbem.github.io/gbem/>
6. Davidson, S., Eaton, B., MacKenzie, L., & Coia, V. (2024). *Stochasim: Channel revegetation and widening*. <https://stochagbem.github.io/stochasim/>
7. Coia, V., & Joe, H. (2021). *Igcop: Computational tools for the IG and IGL copula families*. <https://doi.org/10.32614/CRAN.package.igcop>

Presentations

I delivered the following presentations.

1. Coia, V. (2024). *A machine learning PoF model for hydrotechnical hazards*. Presentation at the 16th Annual Cambio User's Group.
2. Coia, V., & Hairabedian, M. (2023). *A tale of two tributaries: Dependence modelling for hydrologically consistent floodplain maps*. Canadian Water Resources Association - National Conference, Halifax, 2023.
3. Coia, V., & Hille, M. (2023). *Finding the breaking point: What constitutes 'extreme'?* Main stage presentation at BGC Engineering Annual General Meeting.
4. Hairabedian, M., Coia, V., & Grover, P. (2022). *Estimation of the design flood generated by mixed processes for the coldwater river watershed*. Canadian Water Resources Association - National Conference, Canmore, Alberta, 2022.
5. Hairabedian, M., & Coia, V. (2022). *The coldwater river: A watershed moment*. Main stage presentation at BGC Engineering Annual General Meeting.
6. Coia, V., Nolde, N., & Joe, H. (2016). *Forecasting extremes for flooding*. Invited Talk, The 44th Annual Meeting of the Statistical Society of Canada, Brock University, St. Catharines, ON, May 29–June 1, 2016. National.
7. Coia, V., & Jeanniard du Dot, T. (2015). *Using the grammar of graphics and interactivity to explore biologging data in R*. Invited Demonstration, Building a Bioanalytical Theory for Analysis of Marine Mammal Movements: A Peter Wall International Research Roundtable, The University of British Columbia, Vancouver, BC, May 6, 2015. International.
8. Coia, V. (2015). *Flood warning: An application of high-quantile regression*. Contributed Talk, SFU/UBC Joint Graduate Student Seminar (Winter), SFU Harbour Centre, Vancouver, BC, February 28, 2015.
9. Coia, V. (2012). *A new sieve model for extreme values*. Contributed Talk, SFU/UBC Joint Graduate Student Seminar (Fall), SFU Harbour Centre, Vancouver, BC, September 29, 2012.
10. Coia, V., & Huang, M. (2012). *On estimation of heavy tailed distributions*. Contributed Talk, The 40th Annual Meeting of the Statistical Society of Canada, University of Guelph, Guelph, ON, June 3–6, 2012.

I facilitated with the following presentations.

1. Hairabedian, M., Scordo, E., & Coia, V. (2023). *Meta-analysis: Climate change impacts on flooding in british columbia*.

Consulting Reports

1. BGC Engineering Inc. (2024). *Highway 1 chilliwack to hope debris-flow and debris-slide triggering thresholds* (No. 0272107).
2. BGC Engineering Inc. (2024). *Western canadian sedimentary basin - data driven landslide awareness and warning 2024 summary report* (No. P40386).
3. BGC Engineering Inc. (2024). *Erickson creek water balance and michel creek load balance studies* (No. 797037).
4. BGC Engineering Inc. (2023). *Detailed floodplain mapping study: Summary report* (No. 511009).
5. BGC Engineering Inc. (2023). *Extreme weather events and natural disasters – definitions and timing* (No. 1978001).
6. BGC Engineering Inc. (2022). *Frequency-magnitude relationship for the coldwater river (draft report)* (No. 0511009.05.04).

Grants

2025-09 to 2027-09	Open Space Innovation Platform	European Space Agency
	➤ Deciphering compound rain-on-snow flood pathways as observed from satellites.	
2024-09 to 2025-01	The 2024 ISC Grant Program	R Consortium
	➤ Refactoring the distionary R package for submission to CRAN.	
2012-09 to 2015-08	Postgraduate Award (Doctoral, 3-year)	Natural Sciences and Engineering Research Council of Canada
	➤ Doctoral research in dependence modelling and extreme value prediction, with Dr. Harry Joe and Dr. Natalia Nolde.	
2012-09 to 2016-08	Four-Year Fellowship	The University of British Columbia
	➤ Doctoral research in dependence modelling and extreme value prediction, with Dr. Harry Joe and Dr. Natalia Nolde.	
2011-09 to 2012-08	Alexander Graham Bell Canada Graduate Scholarship (Masters)	Natural Science and Engineering Research Council of Canada
	➤ Masters research in the application of heavy-tailed distributions in statistics, with Dr. Mei Ling Huang.	
2010-05 to 2010-08	Undergraduate Student Research Award	Natural Sciences and Engineering Research Council of Canada
	➤ Summer research program in the application of heavy-tailed distributions in statistics, with Dr. Mei Ling Huang.	
2009-05 to 2009-08	Undergraduate Student Research Award	Natural Sciences and Engineering Research Council of Canada
	➤ Summer research program in the application of heavy-tailed distributions in statistics, with Dr. Mei Ling Huang.	

Awards and Honours

2013-06	Governor General of Canada's Gold Medal	Brock University
2012-03-28	RM Davis Surgite Award	Brock University
2011-09	Dean of Graduate Studies Excellence Scholarship	Brock University
2011-06-07	Dean's Gold Medal	Brock University
2011-06-07	Distinguished Undergraduate Student Award in Mathematics	Brock University
2011-03	President's Surgite Award	Brock University

Academic Leadership

- **Co-chair**, Undergraduate Development and Enhancement Committee, Department of Statistics, The University of British Columbia - Vancouver. 2021/2022.

Teaching

Courses taught at The University of British Columbia - Vancouver, BC. I received some of the highest student evaluations in the Faculty of Science in 2021 Winter Term 1 and subsequently formally congratulated by the Dean of Science. My overall median student evaluation score is 4.6 out of 5.

STAT 201 (LEC)	Statistical Inference for Data Science	2021W1
STAT 545A (LEC)	Data Wrangling, Exploration, and Analysis with R - Part I	2021W1
STAT 545B (LEC)	Data Wrangling, Exploration, and Analysis with R - Part II	2021W1
STAT 551 (LEC)	Statistical Consulting Practicum	2021W1
SCIE 300 (LEC)	Communicating Science	2021W1
DSCI 591 (-)	Data Science Internship	2020W2
STAT 201 (LEC)	Statistical Inference for Data Science	2020W1
SCIE 300 (LEC)	Communicating Science	2020W1
STAT 545A (LEC)	Data Wrangling, Exploration, and Analysis with R - Part I	2020W1
STAT 545B (LEC)	Data Wrangling, Exploration, and Analysis with R - Part II	2020W1
DSCI 591 (-)	Data Science Internship	2019W2

STAT 545A (LEC)	Data Wrangling, Exploration, and Analysis with R - Part I	2019W1
DSCI 551 (LEC)	Data Science Workflows	2019W1
DSCI 591 (-)	Data Science Internship	2018W2
BAIT 509 (LEC)	Business Applications of Machine Learning	2018W2
DSCI 554 (LAB)	Experimentation and Causal Inference	2018W2
DSCI 532 (LAB)	Data Visualization II	2018W2
DSCI 562 (LEC, LAB)	Regression II	2018W2
DSCI 561 (LAB)	Regression I	2018W1
DSCI 531 (LEC, LAB)	Data Visualization I	2018W1
DSCI 511 (LEC, LAB)	Programming for Data Science	2018W1
STAT 545A (LEC)	Data Wrangling, Exploration, and Analysis with R - Part I	2018W1
STAT 547M (LEC)	Statistical Computing for Data Science	2018W1
DSCI 591 (-)	Data Science Internship	2017W2
BAIT 509 (LEC)	Business Applications of Machine Learning	2017W2
DSCI 532 (LAB)	Data Visualization II	2017W2
DSCI 562 (LAB)	Regression II	2017W2
DSCI 574 (LAB)	Spatial and Temporal Models	2017W2
DSCI 553 (LAB)	Data Science Workflows	2017W2
STAT 545A (LEC)	Data Wrangling, Exploration, and Analysis with R - Part I	2017W1
STAT 547M (LEC)	Statistical Computing for Data Science	2017W1
DSCI 571 (LAB)	Supervised Learning I	2017W1
DSCI 531 (LEC)	Data Visualization I	2017W1
DSCI 551 (LAB)	Data Science Workflows	2017W1
DSCI 511 (LEC)	Programming for Data Science	2017W1
DSCI 591 (-)	Data Science Internship	2016W2
DSCI 532 (LAB)	Data Visualization II	2016W2
DSCI 562 (LAB)	Regression II	2016W2
DSCI 563 (LAB)	Advanced Machine Learning	2016W2
DSCI 573 (LAB)	Unsupervised Learning	2016W2
DSCI 574 (LAB)	Spatial and Temporal Models	2016W2